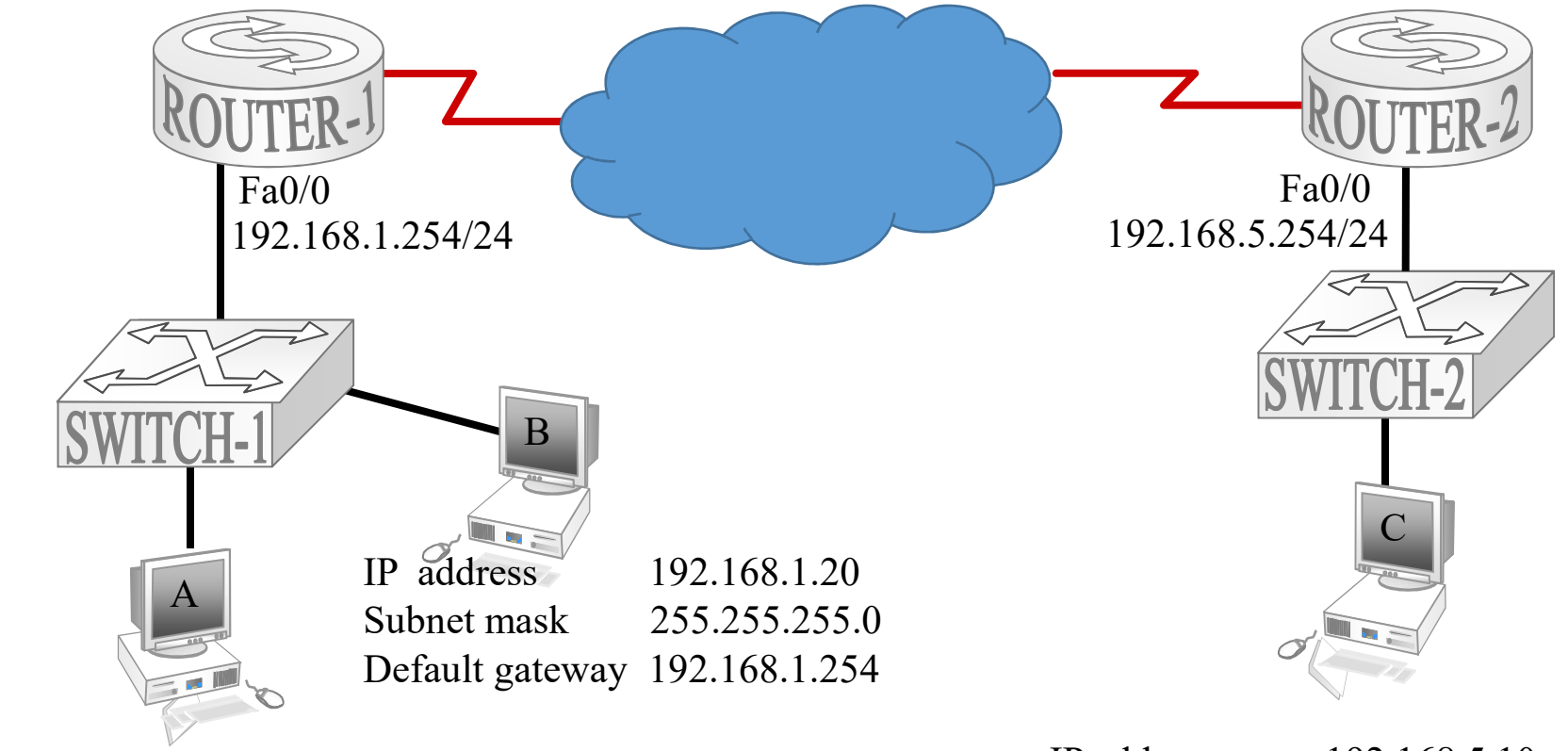
Basic Network Structure



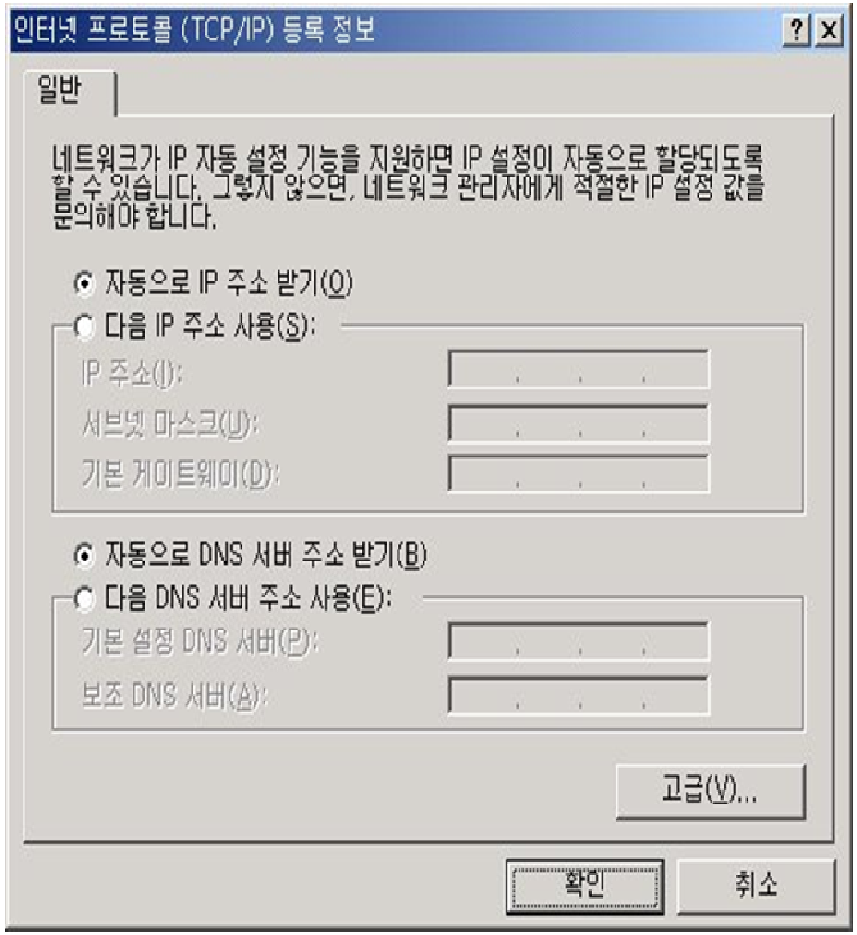
IP address 192.168.1.20
Subnet mask 255.255.255.0
Default gateway 192.168.1.254

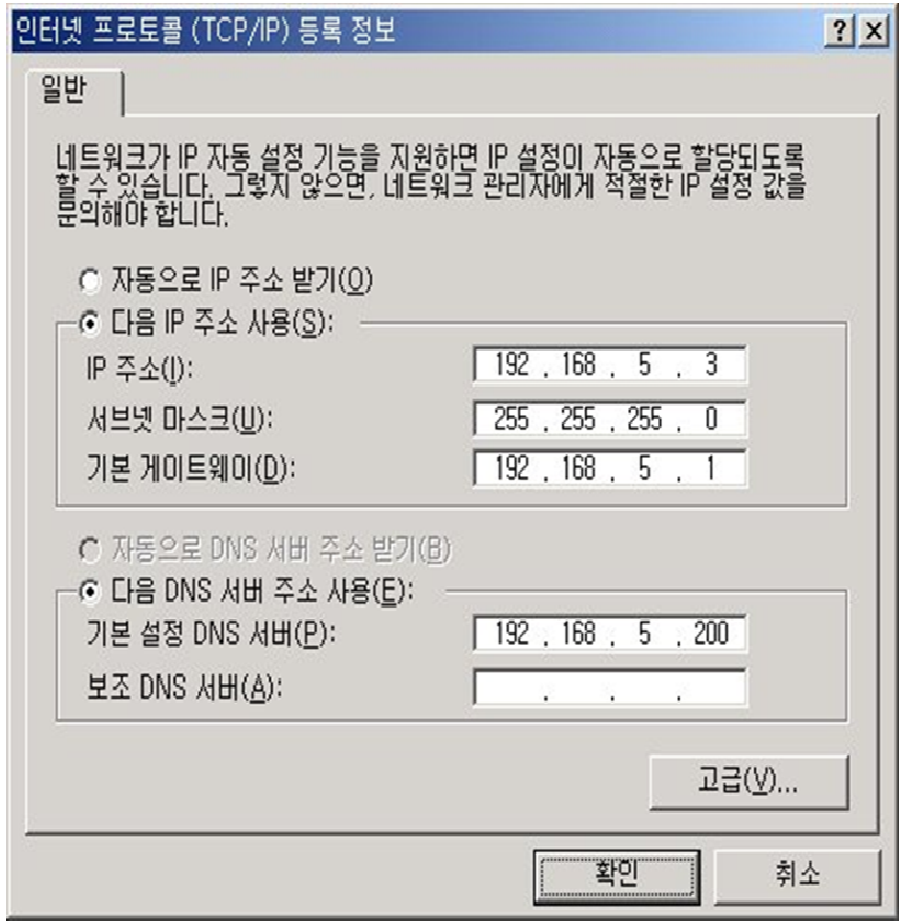
IP address 192.168.1.10
Subnet mask 255.255.255.0
Default gateway 192.168.1.254

IP address 192.168.5.10
Subnet mask 255.255.255.0
Default gateway 192.168.5.254

TCP/IP 등록정보

- 명령어 ipconfig 또는 ipconfig /all





<<유동 IP 환경 >>

<<고정 IP 환경 >>

논리적 주소(3계층주소)

- IP address 구성

  Network ID + Host ID

- Subnet Mask 기능

  - IP address의 Network ID와 Host ID 구분

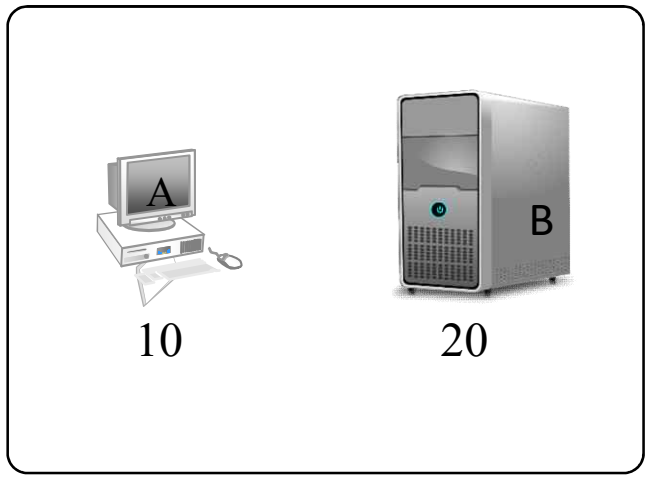
IP address	192.168.1.10
& Subnet mask	& 255.255.255.0
Network ID	192.168.1. 0

255==1
& == X

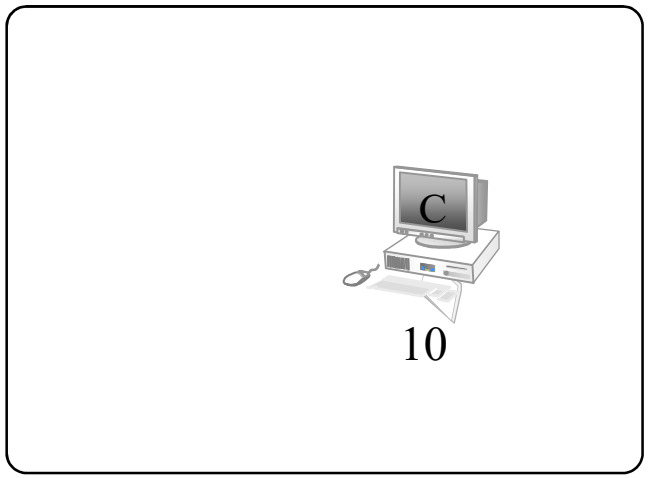
(예) PC A - 192.168.1.10
PC B - 192.168.1.20
PC C - 192.168.5.10

Subnet Mask 255.255.255.0

192.168.1.0

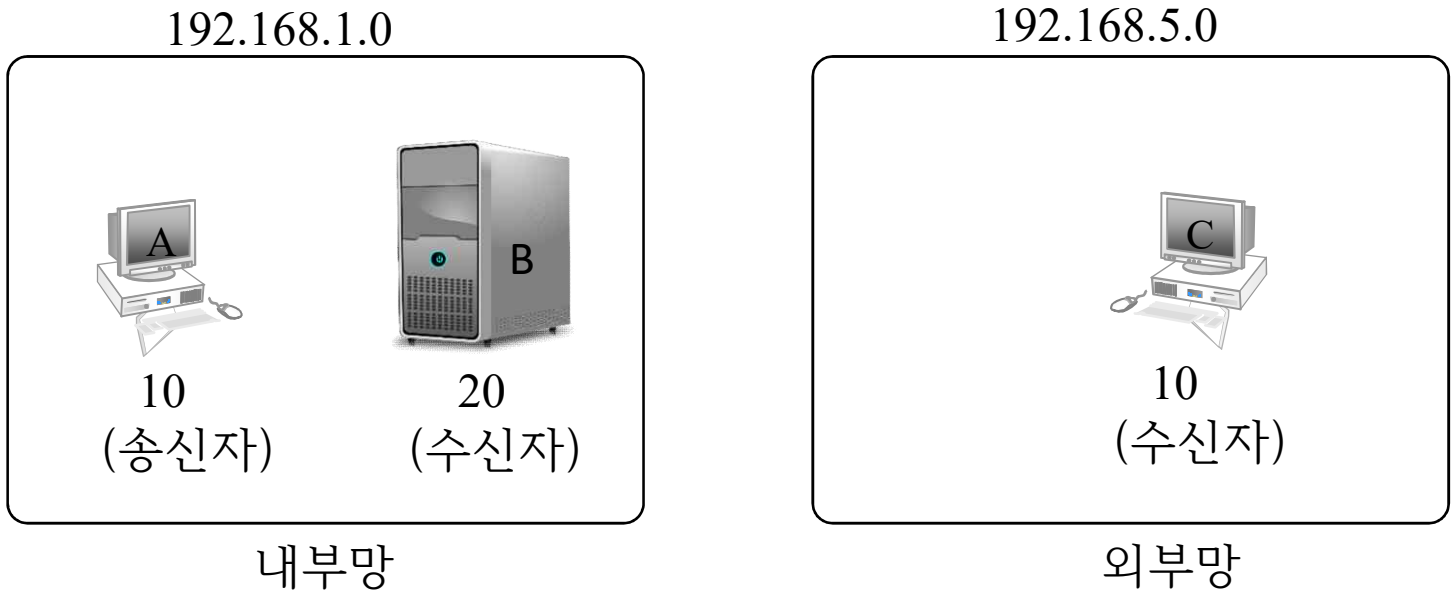


192.168.5.0



내부망과 외부망

- 송신자 : 데이터를 보내는 측 / 수신자 : 데이터를 받는 측

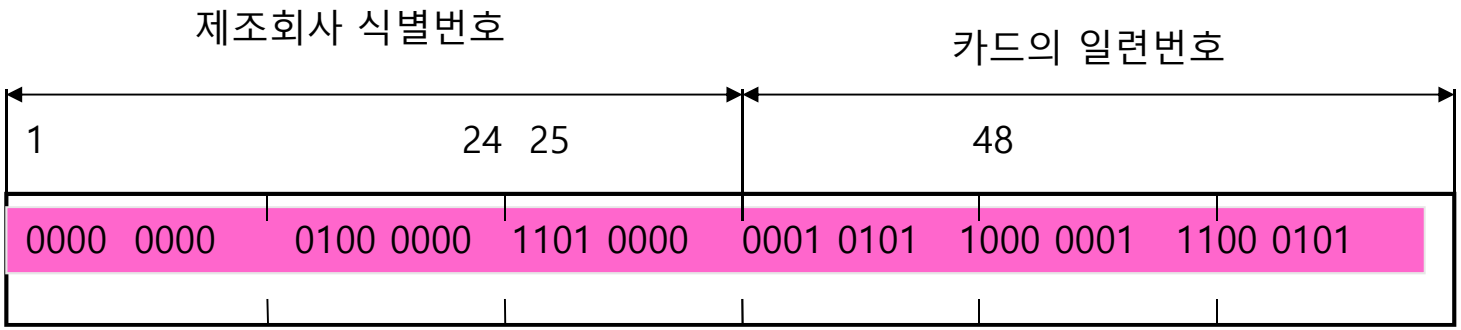


- 내부망 : 송수신자가 동일한 네트워크 ID를 사용
- 외부망 : 송수신지가 서로 다른 네트워크 ID를 사용

물리적 주소(2계층주소)

- Network Interface Card (NIC) 또는 Ethernet Card
- 데이터링크계층의 MAC 계층에 의해 사용되는 48비트의 하드웨어 주소

MAC 주소(16진수 표현) : 00-40-D0-15-81-C5



대표적인 제조회사 식별 번호의 예

- Intel: 00-A0-C9
- 3Com: 00-50-DA
- Realtek: 00-40-D0

MAC Address 확인

- 명령어 ipconfig /all

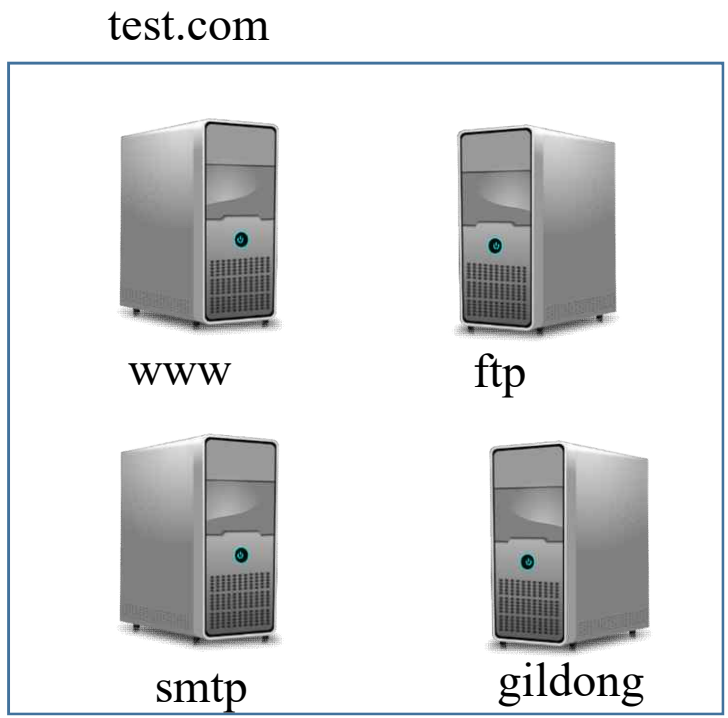
```
무선 LAN 어댑터 Wi-Fi:

   연결별 DNS 접미사. . . . . :
   설명. . . . . . . . . . . . : Intel(R) Dual Band Wireless-AC 8265
   물리적 주소 . . . . . . . . : 68-EC-C5-51-0B-53
   DHCP 사용 . . . . . . . . . : 예
   자동 구성 사용. . . . . . . : 예
   링크-로컬 IPv6 주소 . . . . : fe80::f953:1743:acee:6efa%9(기본 설정)
   IPv4 주소 . . . . . . . . . : 192.168.35.145(기본 설정)
   서브넷 마스크 . . . . . . . : 255.255.255.0
   임대 시작 날짜. . . . . . . : 2020년 6월 25일 목요일 오전 5:41:39
   임대 만료 날짜. . . . . . . : 2020년 6월 25일 목요일 오전 7:18:33
   기본 게이트웨이 . . . . . . : 192.168.35.1
   DHCP 서버 . . . . . . . . . : 192.168.35.1
   DHCPv6 IAID . . . . . . . . : 57208005
   DHCPv6 클라이언트 DUID. . . : 00-01-00-01-24-10-1C-5F-68-EC-C5-51-0B-53
   DNS 서버. . . . . . . . . . : 210.220.163.82
                                 219.250.36.130
   Tcpip를 통한 NetBIOS. . . . : 사용
```

Fully Qualified Domain Name (FQDN,7계층주소)

Host Name + Domain Name

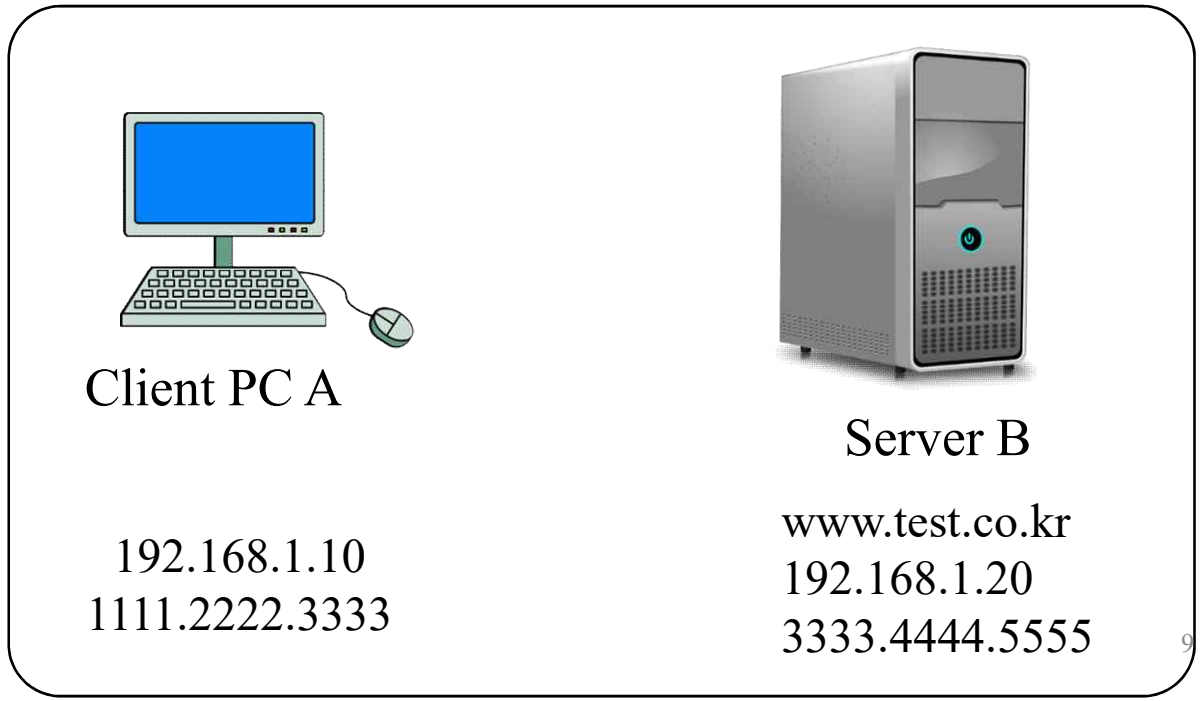
예) www.test.com



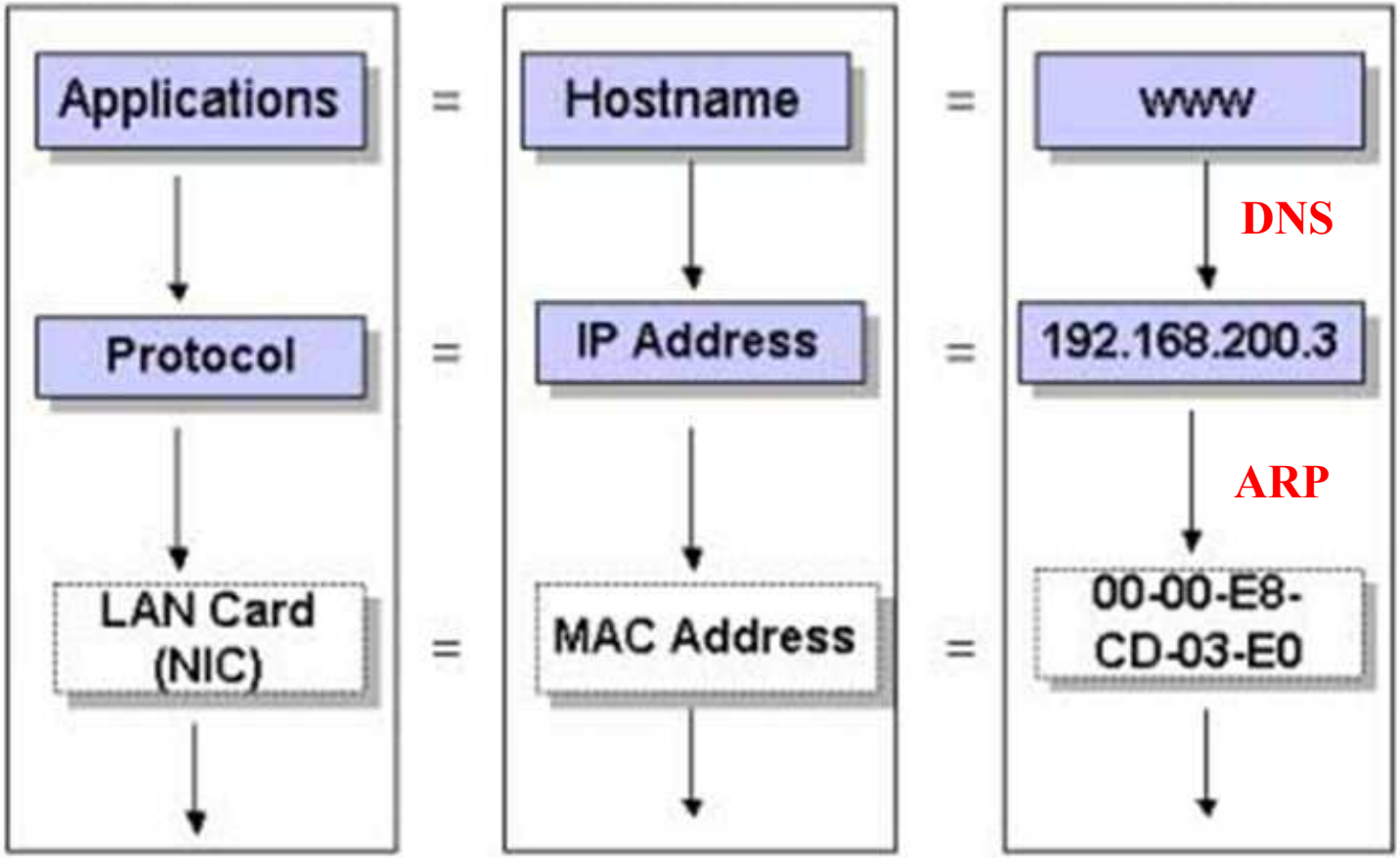
MAC/IP/FQDN Address

- MAC 주소 구성=제조회사+일련번호
- IP 주소 구성=네트워크ID+호스트ID
- FQDN 주소 구성=호스트명+도메인명

그룹주소+고유번호(명)



네트워크 주소와 서비스


Applications
=
Hostname
=
www
DNS
Protocol
=
IP Address
=
192.168.200.3
ARP
LAN Card
(NIC)
=
MAC Address
=
00-00-E8-
CD-03-E0

포트번호

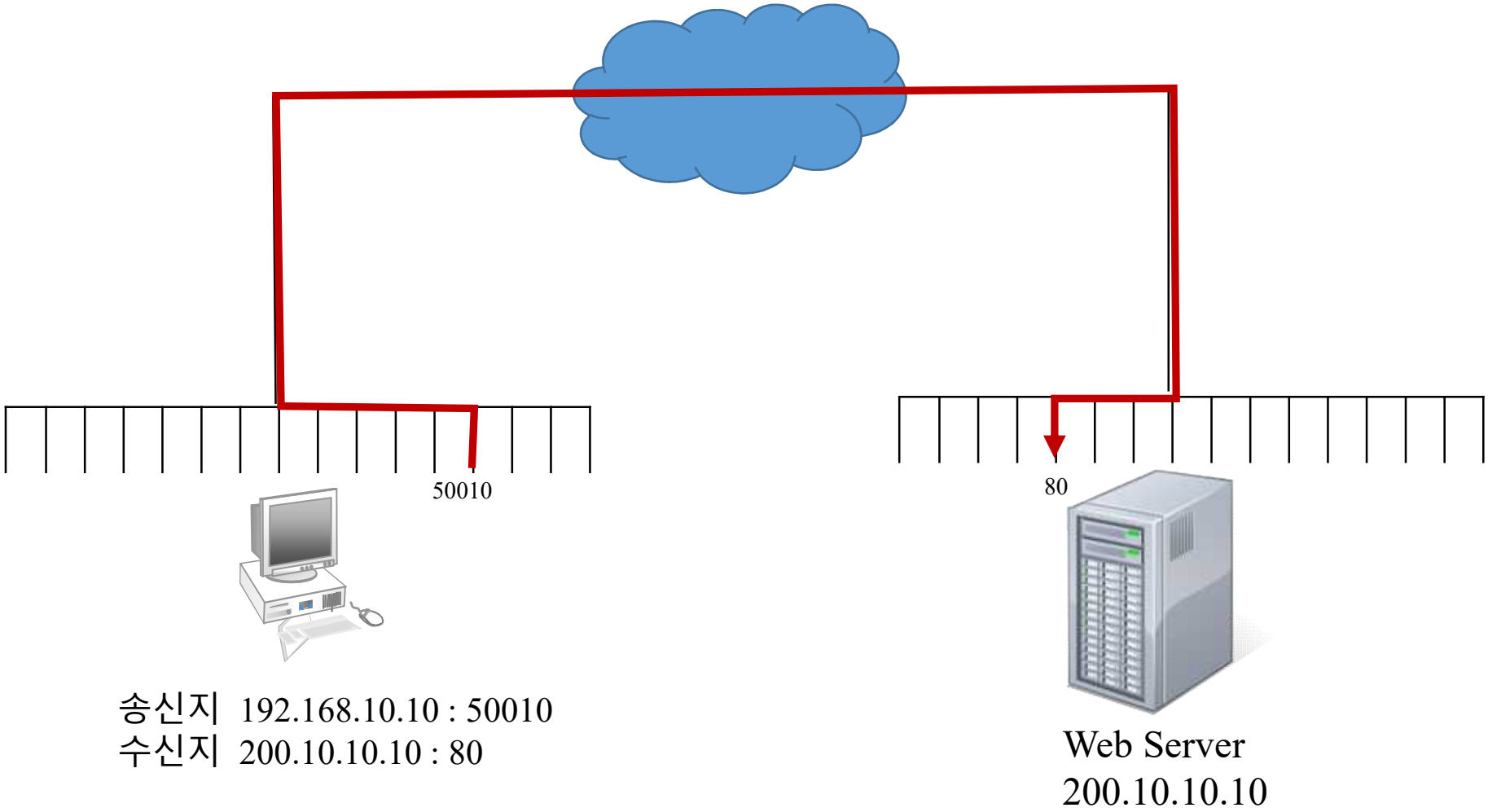
- 데이터 송수신 번호
- 서비스 번호 또는 애플리케이션 번호
- 애플리케이션에서 부착해 전송

  - Well-Known Port : 1-1023

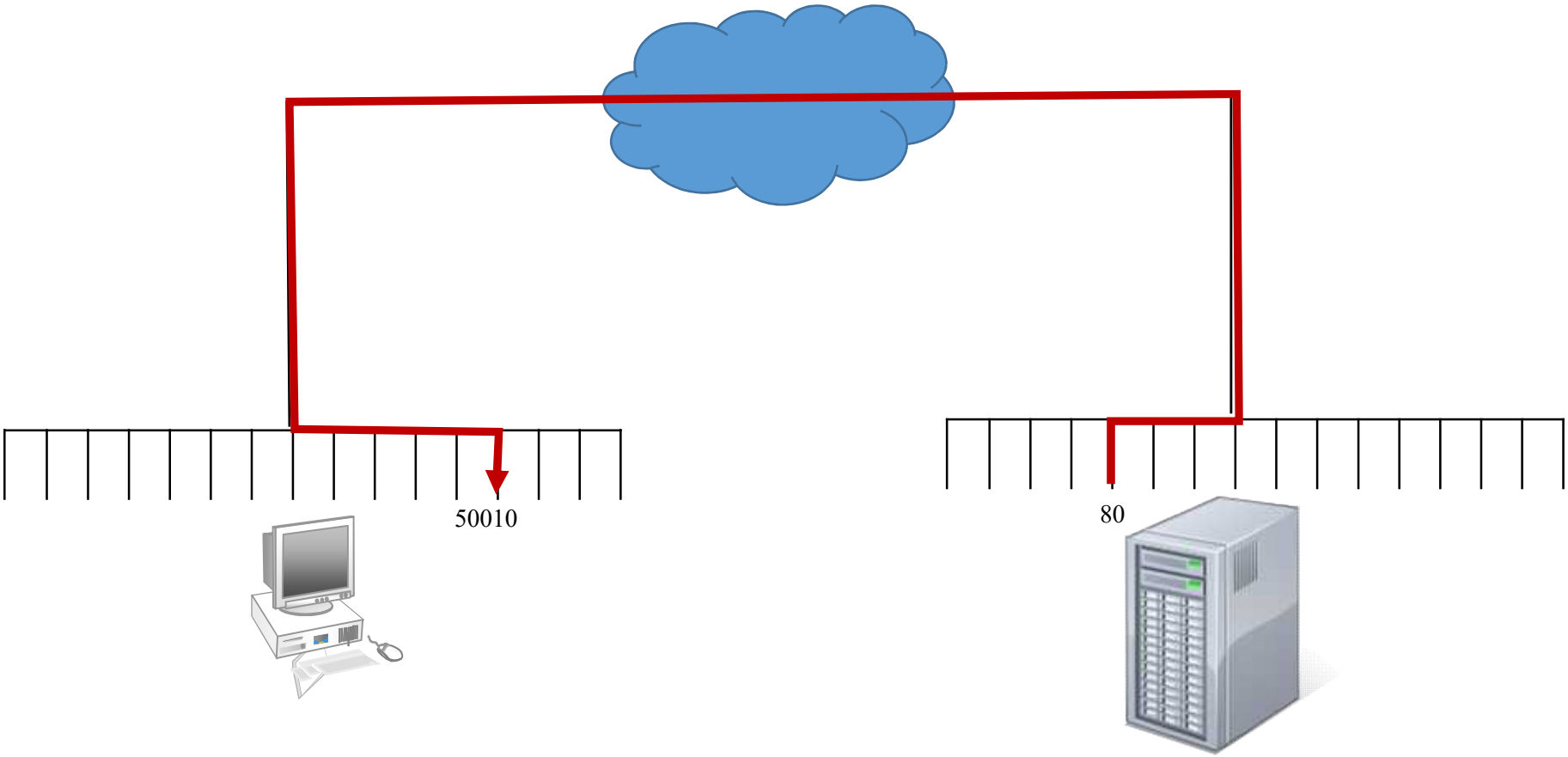
  - Registered Port : 1024-49151

  - Dynamic Port : 49152-65535

포트주소 (송신)


50010
80
송신지 192.168.10.10 : 50010
수신지 200.10.10.10 : 80
Web Server
200.10.10.10

포트주소(수신)



송신지 200.10.10.10 : 80
수신지 192.168.10.10 : 50010

전송형태

Unicast
(1:1)

수신지 MAC	송신지 MAC	송신지 IP	수신지 IP
1234.3456.AB12	67CD.5829.15A7	192.168.1.10	192.168.1.20

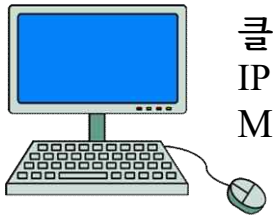
Broadcast
(1:불특정다수)

수신지 MAC	송신지 MAC	송신지 IP	수신지 IP
FFFF.FFFF.FFFF	67CD.5829.15A7	192.168.1.10	255.255.255.255

Multicast
(1:특정다수)

수신지 MAC	송신지 MAC	송신지 IP	수신지 IP
0100.5E0A.8221	67CD.5829.15A7	192.168.1.10	230.10.130.33

• 유니캐스트 전송모드



클라이언트(송신지)
IP 192.168.1.10
MAC 1111.2222.1111

서버(수신지)
IP 192.168.1.20
MAC 1111.2222.2222

L4주소 L3주소 L2주소

전송데이타	수Port	송Port	수신IP	송신 IP	송신지MAC	수신지MAC
get www.test.co.kr	80	50030	192.168.1.20	192.168.1.10	1111.2222.1111	1111.2222.2222

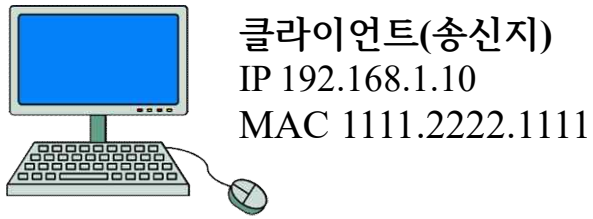
❶ Application ❷ DNS ❸ ARP

브로드캐스트 주소

- 브로드캐스트 IP 주소 : 255.255.255.255(local broadcast)

  192.168.1.255(directed broadcast)

- 브로드캐스트 MAC 주소 : FFFF.FFFF.FFFF



클라이언트(송신지)
IP 192.168.1.10
MAC 1111.2222.1111

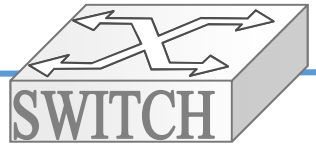
DHCP Server

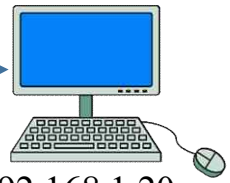
수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
FFFF.FFFF.FFFF	1111.2222.1111	192.168.1.10	255.255.255.255	68	67	DHCP Data

브로드캐스트 전송 예제

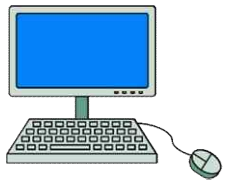
L3주소		L2주소	
수신IP	송신 IP	송신지MAC	수신지MAC
255.255.255.255	192.168.1.10	1111.2222.1111	FFFF.FFFF.FFFF

송신자
IP 192.168.1.10
MAC 1111.2222.1111
BIP 255.255.255.255
BMAC FFFF.FFFF.FFFF

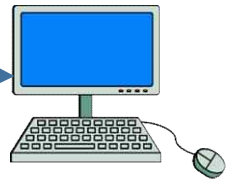




IP 192.168.1.20
MAC 1111.2222.2222
BIP 255.255.255.255
BMAC FFFF.FFFF.FFFF



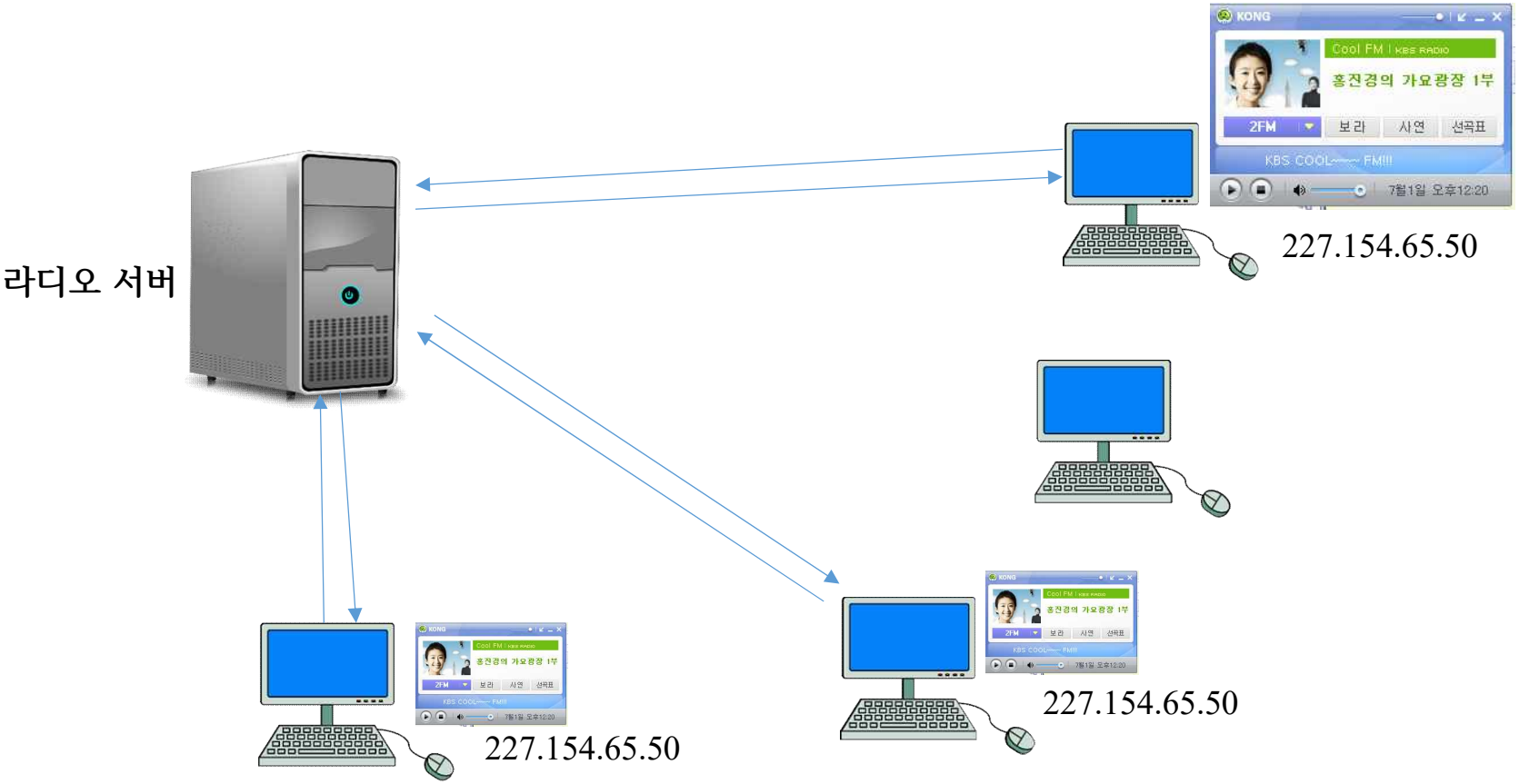
IP 192.168.1.30
MAC 1111.2222.3333
BIP 255.255.255.255
BMAC FFFF.FFFF.FFFF



IP 192.168.1.40
MAC 1111.2222.4444
BIP 255.255.255.255
BMAC FFFF.FFFF.FFFF

멀티캐스트 IP 주소

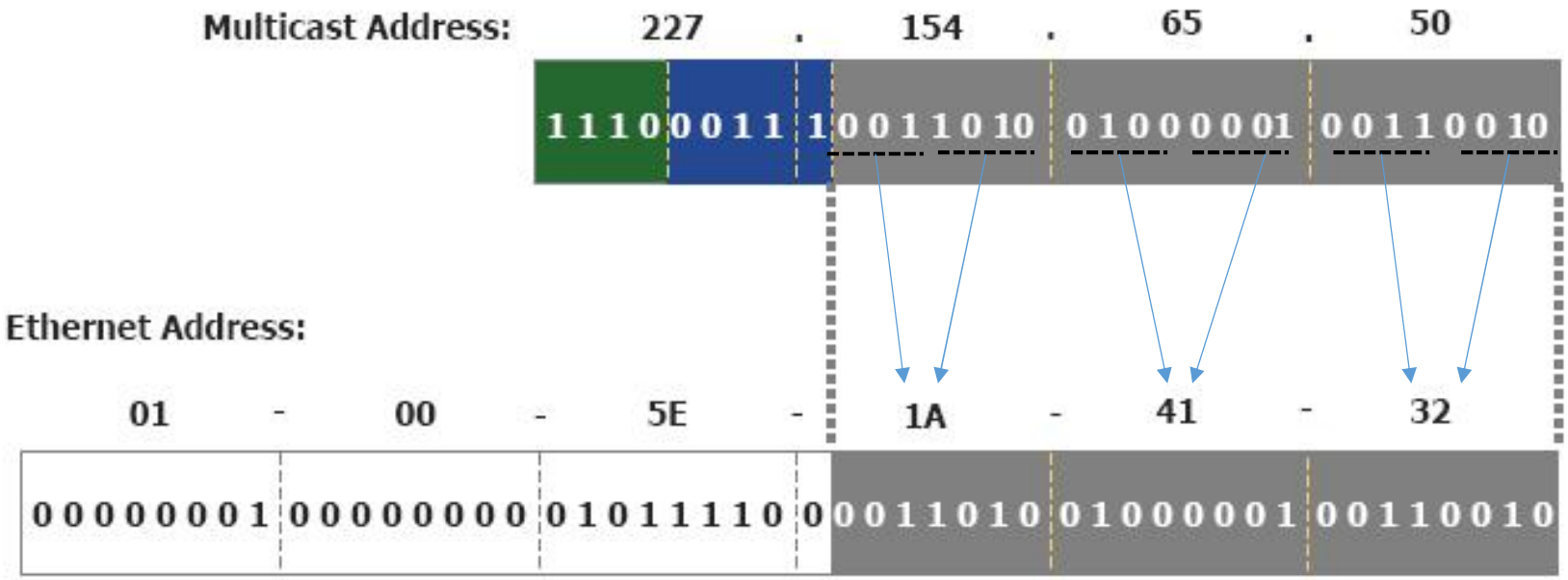
- 멀티캐스트 IP 주소 : 224-239.X.X.X (예) 230.10.10.10



멀티캐스트 MAC 주소

멀티캐스트 MAC 주소 형식 : **<u>0100.5E</u>***XX.XXXX*

- Multicast주소에 대한 MAC address Mapping 적용 예



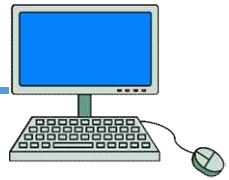
227.154.65.50 ➔ <u>0100.5E</u>1A.4132

멀티캐스트 전송 예제

L3주소		L2주소	
수신IP	송신 IP	송신지MAC	수신지MAC
227.154.65.50	200.10.10.10	2222.4444.6666	0100.5E1A.4132

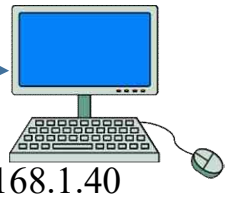
IP 200.10.10.10
MAC 2222.4444.6666
(라디오서버)

IP 192.168.1.10
MAC 1111.2222.1111
Multicast IP 227.154.65.50
Multicast MAC 0100.5E1A.4132

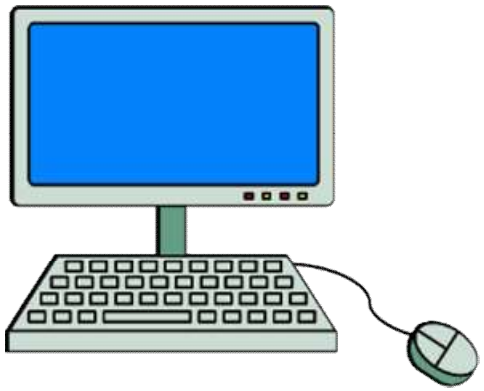


IP 192.168.1.20
MAC 1111.2222.2222

IP 192.168.1.30
MAC 1111.2222.3333
Multicast IP 227.154.65.50
Multicast MAC 0100.5E1A.4132



IP 192.168.1.40
MAC 1111.2222.4444



IP 192.168.1.10
MAC 1111.2222.1111

Broadcast IP 255.255.255.255
Broadcast MAC FFFF.FFFF.FFFF

Multicast IP 227.154.65.50
Multicast MAC 0100.5E1A.4132

통신흐름

1 단계. DNS를 이용하여 수신지 IP 주소 조회

2 단계. 송신자 서브넷 마스크를 이용하여 수신지가 (내부망/외부망)에 존재하는지 확인

3 단계. **수신지 MAC 주소 조회**

- ARP 캐쉬 조회

- ARP Request/Reply 를 이용

4단계. 수신지로 트래픽 전송